VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

B.TECH CSE (AI & ML) · FINAL YEAR PROJECT

Macrograph AI

Multi-agent intelligence for the Indian economy, where every number traces back to an official source

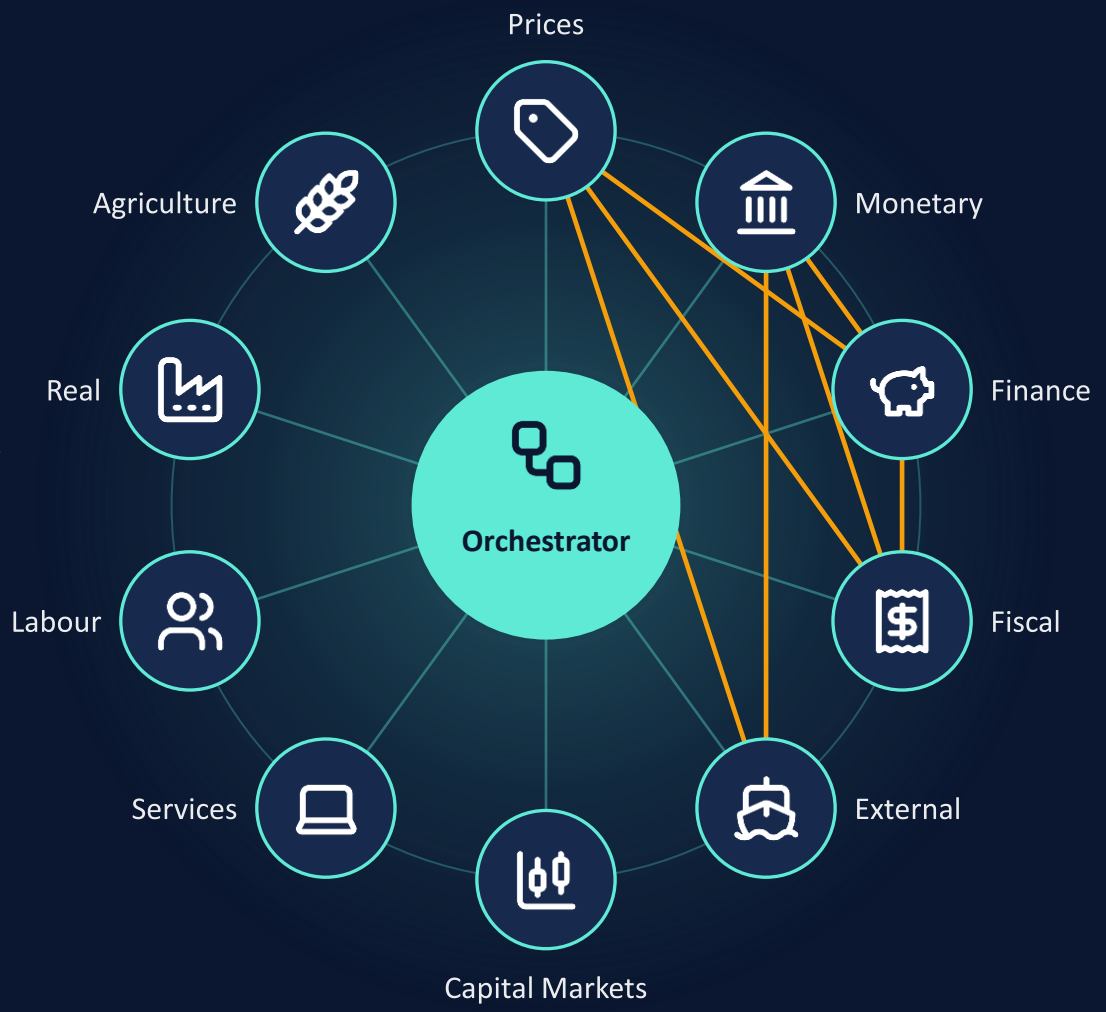
10 sector agents | A2A + FastMCP | Causal knowledge graph

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Guide: [Guide name], [Designation]

School of Computer Science and Engineering, VIT Chennai · November 2026



THE PROBLEM

Scattered sources, invented numbers

A TYPICAL ANALYST QUESTION

“How will a $20 rise in Brent crude travel through India’s inflation, interest rates and growth?”

THE ANSWER NEEDS DATA FROM

- **MoSPI** CPI, GDP, IIP
- **RBI DBIE** repo, credit
- **CGA** deficit, capex
- **DGCI&S** trade
- **NSE / BSE** markets
- **IMD** monsoon
- **Agmarknet** mandi prices
- **EPFO** payroll

Different formats, frequencies, base years and release calendars

Fragmented data

MoSPI, RBI, CGA and DGCI&S publish through APIs, spreadsheets and PDF bulletins, each with its own units, base years and release calendar.

Fluent, unsourced numbers

A language model can quote a CPI figure that appears in no official release; because it looks plausible, it is hard to catch.

Conflicting values

When several agents fetch the same indicator on their own, one report can carry two different vintages of it.

Hidden transmission

Shocks move across sectors with lags and signs; describing each series alone cannot explain how a shock travels.

LITERATURE SURVEY

What prior work leaves open

Work	Approach	Gap for Indian macro answers
Hao & Xie (2025)	Several LLMs act as heterogeneous households	Stylised tasks, no observed data
Lin et al. (2026)	LLM agents simulate macro expectation surveys	Simulated survey answers only
Xiao et al. (2025) TradingAgents	Analyst, researcher and trader agent roles	One shared data feed, no owners
Yang et al. (2020)	Macro knowledge graph mined from literature	Links not re-tested on data
Zhu et al. (2025) KG2RAG	Graph-guided expansion of retrieved chunks	The model still writes numbers
Dadopoulos et al. (2025)	Metadata-driven RAG over financial filings	No claim-level verification
Kato (2026) AI Economist Agent	Reports grounded in executed model runs	U.S. only; no Indian sources

Gaps this project targets

Data ownership across agents

One owner per indicator, enforced by the A2A server

Evidence grades for causal edges

Edge audit labels each edge SEEDED or EMPIRICAL

Machine-checked numbers

Decimal claim verifier against content-addressed evidence

Coverage of India

MoSPI, RBI DBIE, CGA, DGCI&S behind sector FastMCP servers

OBJECTIVES

Seven objectives, one rule

O1

Sector-wise data access

One FastMCP server and local store per sector, fed by official sources

O2

Coordinated multi-agent reasoning

Typed A2A protocol with discovery, routing, loop guards and tracing

O3

Source-grounded answers

Provenance enforced in the protocol; missing data reported, not estimated

O4

Explicit causal model

Knowledge graph of transmission channels with lag, sign and evidence class

O5

Traceable provenance

Source, table, period, retrieval time and hash on every observation

O6

Research instruments

Edge audit, anomaly radar, trajectory, twin, MPC and claim verification

O7

Usable interface

Cited dashboard, streaming chat, causal lab and agent registry

No source, no answer

Every figure carries agent, table and period; gaps are shown as unavailable

PROPOSED SYSTEM

Ten sector agents, one owner per number

Real
GDP, GVA, IIP, GFCF

MoSPI e-Sankhyiki

Agriculture
Crops, MSP, monsoon, mandi prices

Agmarknet · IMD

Prices
CPI and WPI inflation, food, fuel and core

MoSPI e-Sankhyiki

Monetary
Repo corridor, M3, liquidity, stance

RBI DBIE

Finance
Bank credit, GNPA, WALR and MCLR

RBI DBIE tables

Capital Markets
NIFTY, India VIX, FII/DII flows, G-Sec

yfinance · SEBI

Fiscal
Fiscal deficit, GST, capex, public debt

CGA · Union Budget

External
Forex reserves, trade, BoP, USD/INR

RBI · DGCI&S

Labour
PLFS unemployment, EPFO payroll

MoSPI PLFS · EPFO

Services
Services GVA, PMI, IT exports

MoSPI · PMI releases

A2A for reasoning
agents talk only through typed requests

MCP for data
each agent reads only its own MCP server

No shortcuts
sector packages never import each other

ARCHITECTURE

Reasoning flows apart from data



ORCHESTRATION

From question to cited report in four steps

1 Route

`decompose`

Scores Agent Cards against the query; an LLM chooses only if no card matches

2 Delegate

`parallel_a2a_execute`

Concurrent sector_analysis requests, plus one bounded follow-up round

3 Enrich

`kg_causal_enrichment`

Shortest causal paths between observed indicators; optional shock simulation

4 Synthesise

`synthesis`

Indicator matrix with provenance hashes, causal diagram and source table

WORKED RUN · OCTOBER 2026

"How does a rise in crude oil prices affect inflation and RBI repo rate policy?"

Shock: Brent +$20 / bbl

Routed to Monetary, Prices and Agriculture

OBSERVATIONS RETURNED

Indicator	Value	Period	Source
CPI headline (YoY)	**5.54%**	Sep 2026	MoSPI
Repo rate	**5.25%**	Feb 2026	RBI DBIE
M3 growth (YoY)	**unavailable**	Jun 2026	RBI DBIE

Every value keeps its period, status and SHA-256 hash

GRAPH PATHS FOUND

CPI → repo · 1 hop, 1 month
Brent → CPI · 2 hops, 3 months
Brent → repo · 3 hops, 4 months

≈ 11 s

end to end with live official data and no LLM key

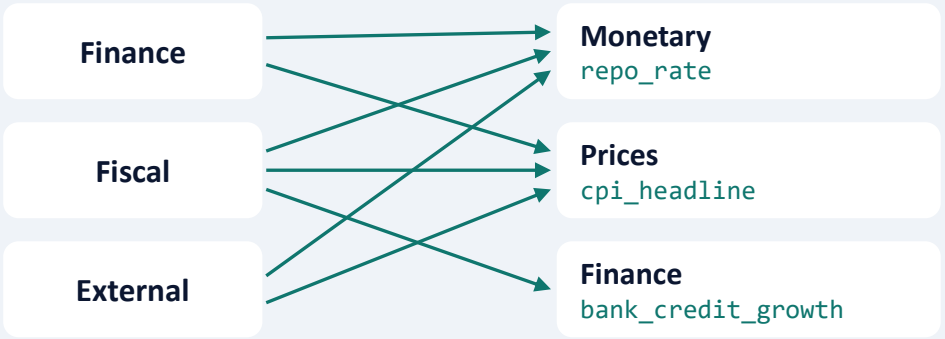
A protocol that refuses unsourced answers

Six checks on every A2A request

1. **Loop guards**
   Depth ≤ 5 · at most 25 requests per conversation · no cycles
2. **Declared dependency**
   Sector-to-sector calls only on the 7 declared edges
3. **Capability**
   The task must be listed in the receiver's Agent Card
4. **Schema**
   Strict Pydantic v2 envelopes; extra fields rejected
5. **Timeout and retry**
   90 s request, 25 s peer; only transient errors are retried
6. **Provenance** — **No Source, No Answer**
   A successful reply with no sources is rejected

7 declared peer edges

consumer → provider · A2A task



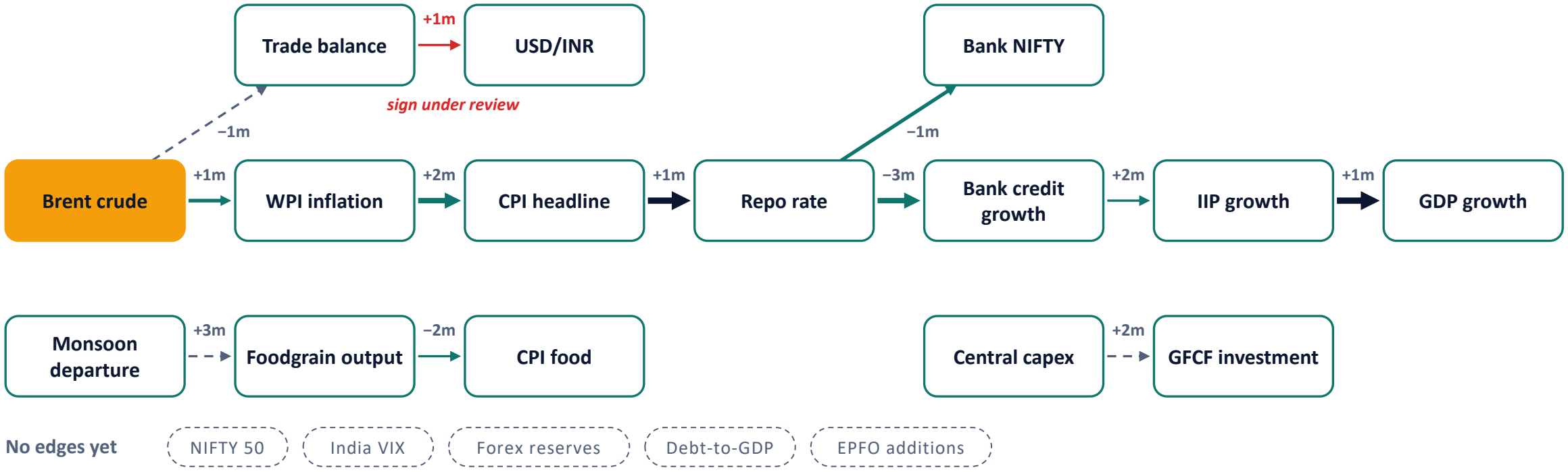
MCP data path for every value

Live official source → Cached snapshot → Structured unavailable

Never an estimate. Each value keeps source URL, dataset, table, period, retrieval time and a record reference.

CAUSAL KNOWLEDGE GRAPH

20 indicators, 12 typed transmission edges



No edges yet

NIFTY 50 · India VIX · Forex reserves · Debt-to-GDP · EPFO additions

EVIDENCE TIER, AS TYPED IN THE ONTOLOGY (PRIORS)

- Theory
- Statistical association
- Lagged relationship
- Granger predictive
- Structural causal model

EDGE ANATOMY

Each edge: stable rel-id, lag, sign, confidence and mechanism
Checked on load: no self-loops, unknown endpoints, duplicates
NetworkX answers path and impact queries; Neo4j mirror optional

SCENARIO ENGINE

Tracing a $20 oil shock, hop by hop

PROPAGATION RULE

Δ target = shock × Π (sign × 0.55 × confidence) × 0.15

Lags add up along the path, confidence multiplies, and the band is ±25% of Δ.

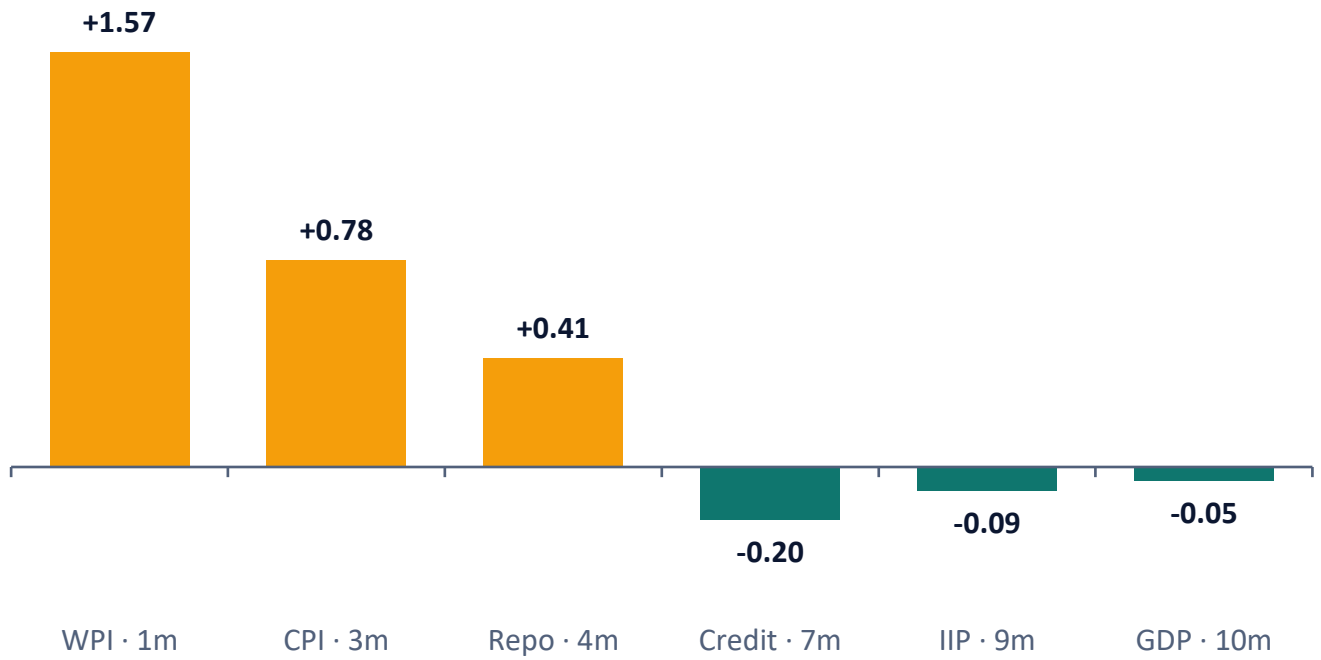
Hop 1: 20 × (0.55 × 0.95) × 0.15 = +1.57
so WPI moves from 2.45 to 4.02

SIMULATED

A transparent heuristic, not a VAR: the constants are design choices and deltas stay in each indicator's own unit.

USD/INR falls after the oil shock, which is implausible: the run exposed a doubtful + sign on trade balance → USD/INR.

Simulated change along the main chain, percentage points (Brent +$20/bbl)



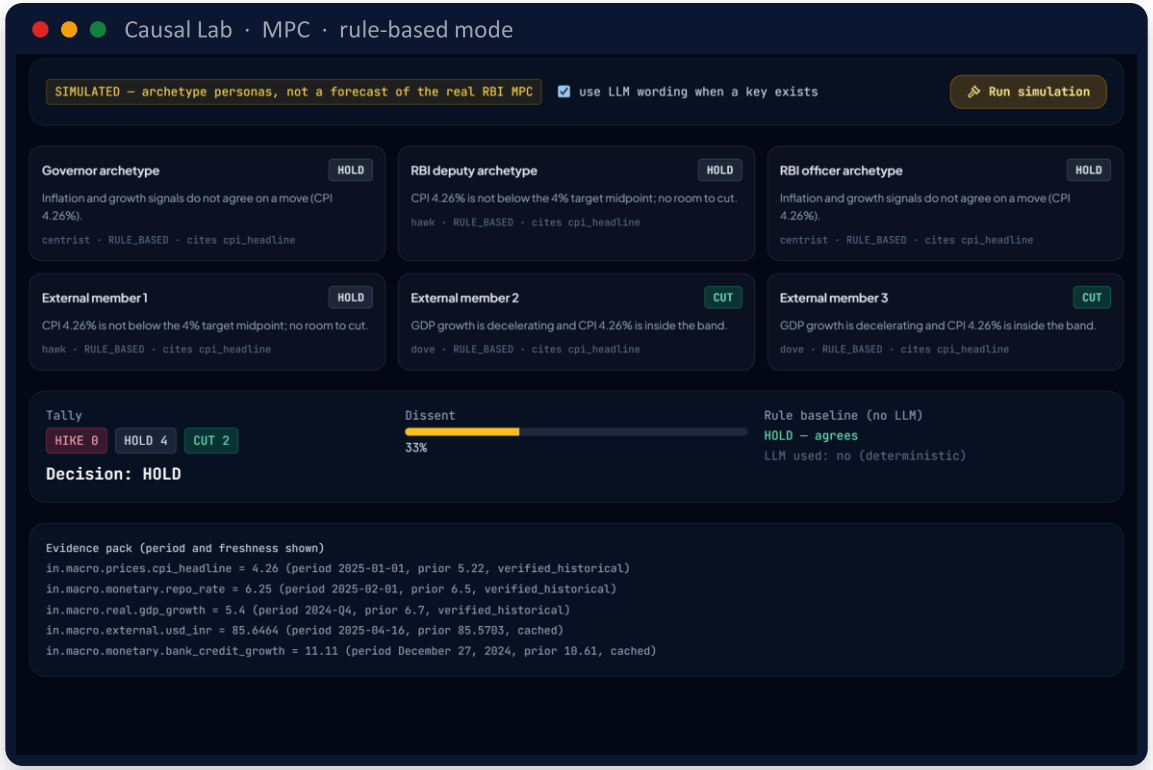
Path confidence 0.95 → 0.85 → 0.81 → 0.71 → 0.59 → 0.57 over six hops

Live baselines live CPI 5.54% (MoSPI, Sep 2026) replaces the default; peak 6.32%

CAUSAL LAB

A lab that tests its own assumptions



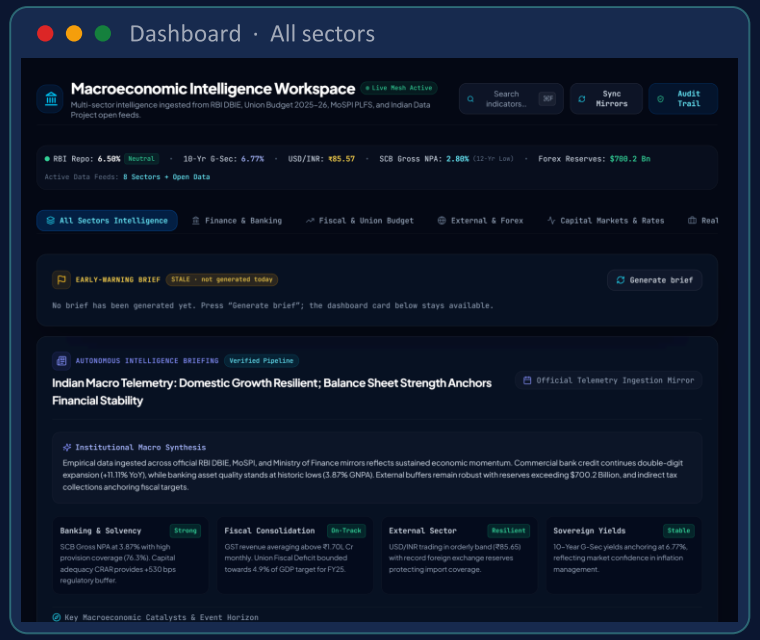


- **Path** shortest route and cumulative lag
- **Shockwave** month-by-month propagation, step or decay
- **Twin** base vs shocked world, weakest hop

- **MPC** six simulated seats vs a rule baseline
- **Radar** z-score anomalies, propagating or isolated
- **Edge audit** Granger re-test: SEEDED or EMPIRICAL

PRODUCT SHOWCASE

A working product, not a notebook



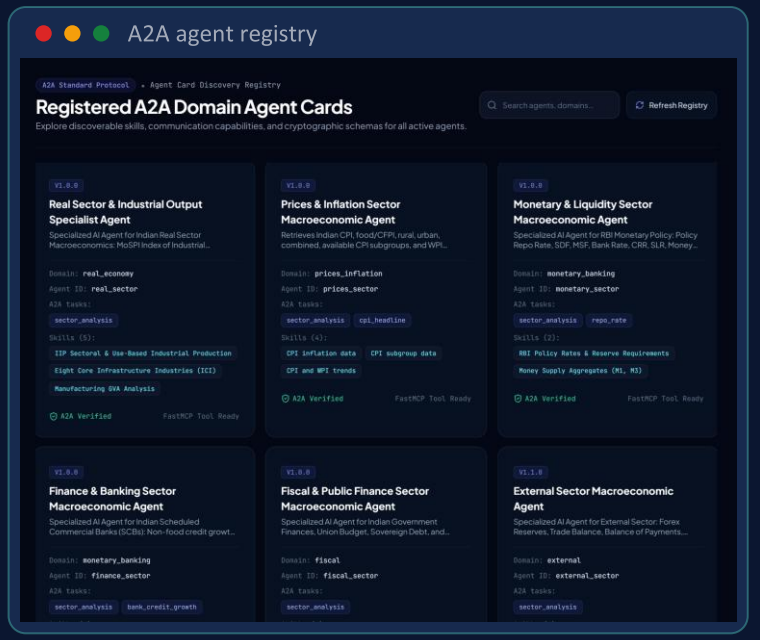
Cross-sector dashboard

Live ticker, early-warning brief and sector tabs; KPIs carry their table and period.



Finance and banking

Credit, asset quality and capital gauges from RBI DBIE tables r539, r330, r531 and r689.



A2A agent registry

Agent Cards with skills, A2A tasks and the declared peer edges between sectors.

ALSO IN THE APP Streaming chat with a live A2A trace · Evidence drawer · Knowledge-graph explorer · Google sign-in · Command palette

RESULTS

What the measurements show

550 / 551

test cases passed across all ten sector suites (455 test functions)

PYTEST · 12 OCT 2026

≈ 11 s

cross-sector answer with live official data and no LLM key

END-TO-END RUN

48.9k

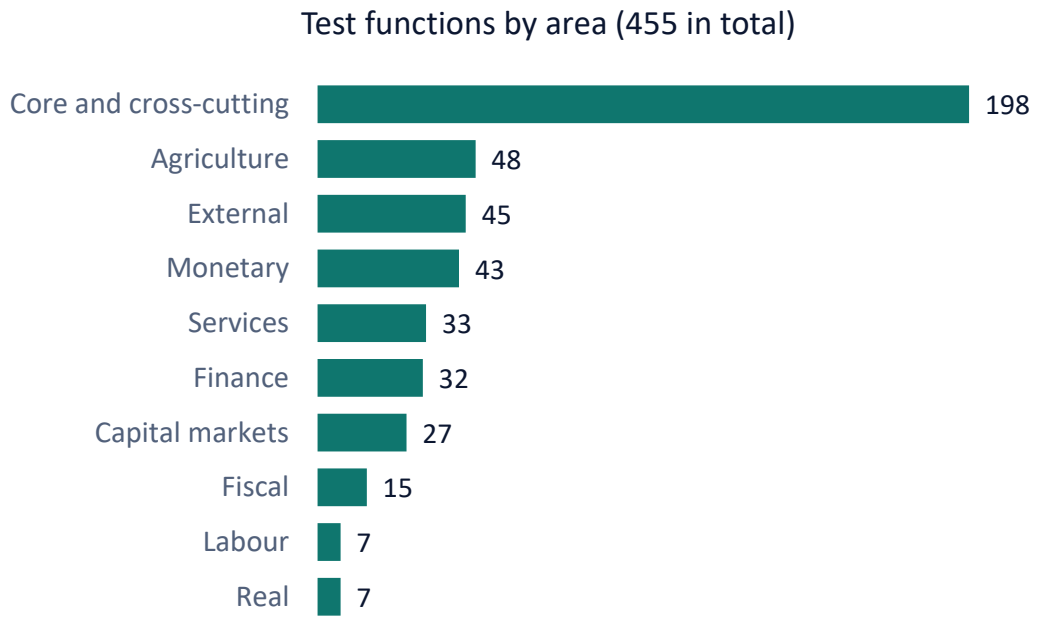
lines of code, excluding tests:
37.6k Python and
11.3k TypeScript

REPOSITORY COUNT

0 / 12

causal edges testable today: series overlap at most 2 of the 10 months required

EDGE EVIDENCE AUDIT



Anomaly radar

18 series scanned, 8 long enough
Trade balance, Aug 2024: z = −2.65
classified as isolated

Simulated MPC

HOLD by 4 to 2, dissent 0.33
matches the rule-only baseline

Honest limits, concrete next steps

Known limitations

Seeded edge statistics
Confidences and p-values are curated priors; none testable yet

Heuristic scenario engine
Fixed 0.55 and 0.15 constants, unit-blind deltas, band is not a CI

Verifier not in the answer path
Claim verification is tested, but no live answer calls it yet

Short histories
10 of 18 radar series have fewer than 8 observations

No documentary retrieval
RBI minutes and the Economic Survey are not indexed

Open cost-bearing endpoints
Chat, brief and MPC routes lack auth and rate limits

Also logged from the live run: "None%" shown for M3, permissive agriculture routing, a misleading no-key log line.

Twelve-week roadmap

WEEKS 1–2 Credibility fixes
Wire claim verification into synthesis, fix the edge sign, surface silent failures, repair the failing test

WEEKS 3–6 Causal modelling
Estimate edges from long series (ADF, BH-corrected Granger, local projections); back-tested LP or BVAR scenarios

WEEKS 5–8 Evaluation
About 250 questions with gold answers, five baselines and ablations; point-in-time MPC benchmark

WEEKS 9–12 Consolidation
Results analysis, endpoint hardening, documentation and a workshop paper draft

CONCLUSION

Every number has an owner and a source

O1 Data access DELIVERED	O2 Coordination DELIVERED	O3 Grounded answers PARTLY	O4 Causal model PARTLY	O5 Provenance DELIVERED	O6 Research tools PROTOTYPES	O7 Interface DELIVERED

Responsibility for each number lies with one agent, every number travels with its source, and missing evidence is reported openly, while the platform's own audit shows how much of the causal reasoning still rests on assumptions.

NEXT

1. Verify every claim before it is shown
2. Estimate edges from long series
3. Benchmark against five baselines

Thank you

Questions and discussion welcome

github.com/Hariprasaadh/macrograph-ai

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